PHYS4055 Week 6 Tutorial Questions

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Solution to Question 1

1. From the definitions of the symmetric and antisymmetric states,

$$|S\rangle + |A\rangle = \sqrt{2}|e,g\rangle.$$

Therefore,

$$\boxed{|e,g\rangle = \frac{|S\rangle + |A\rangle}{\sqrt{2}}}$$

and the complete initial state is

$$|\Psi(0)\rangle = \frac{1}{\sqrt{2}}\left(|S;0\rangle + |A;0\rangle\right).$$

2. For the symmetric state,

$$\begin{aligned}\hat{J}_-|S\rangle &= \frac{1}{\sqrt{2}}\left(\hat{\sigma}_-^{(1)} + \hat{\sigma}_-^{(2)}\right)\left(|e,g\rangle + |g,e\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(|g,g\rangle + |g,g\rangle\right)\\ &= \sqrt{2}|g,g\rangle.\end{aligned}$$

Hence,

$$\boxed{\hat{J}_-|S\rangle = \sqrt{2}|g,g\rangle.}$$

For the antisymmetric state,

$$\begin{aligned}\hat{J}_-|A\rangle &= \frac{1}{\sqrt{2}}\left(\hat{\sigma}_-^{(1)} + \hat{\sigma}_-^{(2)}\right)\left(|e,g\rangle - |g,e\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(|g,g\rangle - |g,g\rangle\right)\\ &= 0.\end{aligned}$$

Thus,

$$\boxed{\hat{J}_-|A\rangle = 0.}$$

3. For the symmetric state, the two possible atomic emission amplitudes add together because they produce the same final state. This gives an enhanced coupling of $\sqrt{2}g$.

For the antisymmetric state, the two amplitudes have opposite signs and cancel. Therefore, the antisymmetric state does not couple to the cavity mode.

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4. The initial state contains equal symmetric and antisymmetric components. The symmetric component evolves while the antisymmetric component remains unchanged. Therefore,

$$\boxed{|\Psi(t)\rangle = \frac{1}{\sqrt{2}}\left[\cos(\sqrt{2}gt)|S;0\rangle + |A;0\rangle - i\sin(\sqrt{2}gt)|g,g;1\rangle\right].}$$

Returning to the individual-atom basis gives

$$\boxed{\begin{aligned}|\Psi(t)\rangle = &\frac{1+\cos(\sqrt{2}gt)}{2}|e,g;0\rangle \\ &+ \frac{\cos(\sqrt{2}gt)-1}{2}|g,e;0\rangle \\ &- \frac{i\sin(\sqrt{2}gt)}{\sqrt{2}}|g,g;1\rangle.\end{aligned}}$$

5. Only the state $|g,g;1\rangle$ contains a cavity photon. Therefore,

$$\begin{aligned}\langle\hat{n}_{\text{cav}}\rangle &= \left|-\frac{i\sin(\sqrt{2}gt)}{\sqrt{2}}\right|^2 \\ &= \boxed{\frac{1}{2}\sin^2(\sqrt{2}gt)}.\end{aligned}$$

Both $|S;0\rangle$ and $|A;0\rangle$ contain one excited atom. Their probabilities are respectively

$$\frac{1}{2}\cos^2(\sqrt{2}gt) \qquad \text{and} \qquad \frac{1}{2}.$$

Consequently,

$$\begin{aligned}\langle\hat{N}_{\text{at}}\rangle &= \frac{1}{2}\cos^2(\sqrt{2}gt) + \frac{1}{2} \\ &= \boxed{1 - \frac{1}{2}\sin^2(\sqrt{2}gt)}.\end{aligned}$$

The expected total number of excitations is

$$\begin{aligned}\langle\hat{N}_{\text{total}}\rangle &= \langle\hat{N}_{\text{at}}\rangle + \langle\hat{n}_{\text{cav}}\rangle \\ &= 1 - \frac{1}{2}\sin^2(\sqrt{2}gt) + \frac{1}{2}\sin^2(\sqrt{2}gt) \\ &= 1.\end{aligned}$$

Thus,

$$\boxed{\langle\hat{N}_{\text{total}}\rangle = 1,}$$

so the total number of excitations is conserved.

6. The probability of finding one photon in the cavity is

$$P_{\text{cav}}(t) = \frac{1}{2}\sin^2(\sqrt{2}gt).$$

This is largest when

$$\sin^2(\sqrt{2}gt) = 1.$$

The first such time is

$$\boxed{t = \frac{\pi}{2\sqrt{2}g}.}$$

The maximum probability is

$$\boxed{P_{\text{cav}}^{\max} = \frac{1}{2}.}$$

The initial state contains equal symmetric and antisymmetric components. The symmetric component can transfer its excitation to the cavity, but the antisymmetric component does not interact with the cavity. Since half of the initial state remains in the antisymmetric component, the cavity photon probability cannot exceed 1/2.

7. From the state in the individual-atom basis, the probability that atom 1 is excited is

$$\boxed{P_{e,1}(t) = \frac{\left[1 + \cos(\sqrt{2}gt)\right]^2}{4}.}$$

Similarly, the probability that atom 2 is excited is

$$\boxed{P_{e,2}(t) = \frac{\left[1 - \cos(\sqrt{2}gt)\right]^2}{4}.}$$

Complete transfer from atom 1 to atom 2 occurs when

$$\cos(\sqrt{2}gt) = -1.$$

The first such time is

$$\boxed{t = \frac{\pi}{\sqrt{2}g}.}$$

At this time,

$$P_{e,1} = 0, \qquad P_{e,2} = 1, \qquad P_{\text{cav}} = 0.$$

The complete state is

$$|\Psi(t)\rangle = -|g, e; 0\rangle.$$

The minus sign is an overall phase and has no physical consequence. Therefore, the excitation has been transferred completely from atom 1 to atom 2.

Solution to Question 2

1. The initial overlap is

$$\begin{aligned}\langle\psi_+|\psi_-\rangle &= \frac{1}{5}\left(2\langle 0| + \langle 1|\right)\left(2|0\rangle - |1\rangle\right) \\ &= \frac{4-1}{5} \\ &= \boxed{\frac{3}{5}}.\end{aligned}$$

Since the overlap is non-zero, the states are not orthogonal and cannot be distinguished perfectly by a deterministic measurement.

2. Applying the successful conditional operation gives

$$\begin{aligned}\hat{K}_{\mathrm{s}}|\psi_{\pm}\rangle &= \left(\frac{1}{2}|0\rangle\langle 0| + |1\rangle\langle 1|\right)\frac{2|0\rangle \pm |1\rangle}{\sqrt{5}} \\ &= \boxed{\frac{|0\rangle \pm |1\rangle}{\sqrt{5}}}.\end{aligned}$$

   These states are unnormalised.

3. The probability of success is the squared norm of the unnormalised output:

$$\begin{aligned}P_{\mathrm{s},\pm} &= \langle\psi_{\pm}|\hat{K}_{\mathrm{s}}^{\dagger}\hat{K}_{\mathrm{s}}|\psi_{\pm}\rangle \\ &= \left\|\frac{|0\rangle \pm |1\rangle}{\sqrt{5}}\right\|^2 \\ &= \frac{1+1}{5} \\ &= \boxed{\frac{2}{5}}.\end{aligned}$$

   The probability of success is the same for both input states.

4. Dividing each successful output by the square root of its success probability gives

$$|\phi_{\pm}\rangle = \frac{\hat{K}_{\mathrm{s}}|\psi_{\pm}\rangle}{\sqrt{P_{\mathrm{s},\pm}}} = \boxed{\frac{|0\rangle \pm |1\rangle}{\sqrt{2}}}.$$

   Their overlap is

$$\begin{aligned}\langle\phi_{+}|\phi_{-}\rangle &= \frac{1}{2}\left(\langle 0| + \langle 1|\right)\left(|0\rangle - |1\rangle\right) \\ &= \frac{1-1}{2} \\ &= \boxed{0}.\end{aligned}$$

   The successful output states are therefore orthogonal.

5. The operation associated with $\hat{K}_{\mathrm{s}}$ is conditional and is not a deterministic, trace-preserving operation. It produces the orthogonal output states only when the successful detector result occurs, which has probability 2/5.

   The complete physical measurement must include both the successful and failed outcomes. Indeed,

$$\begin{aligned}\hat{K}_{\mathrm{s}}^{\dagger}\hat{K}_{\mathrm{s}} &= \frac{1}{4}|0\rangle\langle 0| + |1\rangle\langle 1|, \\ \hat{K}_{\mathrm{f}}^{\dagger}\hat{K}_{\mathrm{f}} &= \frac{3}{4}|0\rangle\langle 0|,\end{aligned}$$

   and hence

$$\hat{K}_{\mathrm{s}}^{\dagger}\hat{K}_{\mathrm{s}} + \hat{K}_{\mathrm{f}}^{\dagger}\hat{K}_{\mathrm{f}} = |0\rangle\langle 0| + |1\rangle\langle 1| = I \quad \text{within the subspace spanned by } \{|0\rangle, |1\rangle\}.$$

   Thus, the increased distinguishability of the successful outputs is accompanied by a non-zero probability of failure.

6. Applying the failure operator gives

$$\begin{aligned}\hat{K}_{\mathrm{f}}|\psi_{\pm}\rangle &= \frac{\sqrt{3}}{2}|0\rangle\langle 0|\frac{2|0\rangle \pm |1\rangle}{\sqrt{5}} \\ &= \boxed{\sqrt{\frac{3}{5}}|0\rangle}.\end{aligned}$$

The probability of this result is

$$P_{\mathrm{f},\pm} = \left\| \sqrt{\frac{3}{5}}|0\rangle \right\|^2 = \boxed{\frac{3}{5}}.$$

After normalisation, both input states produce the same conditional output:

$$\boxed{|\phi_{\mathrm{f},\pm}\rangle = |0\rangle.}$$

The failed result therefore provides no information about whether the input was $|\psi_+\rangle$ or $|\psi_-\rangle$.